

Derivative of Exponential Function

1. Differentiate:

- a. e^{2x}
- b. e^{-3x}
- c. $-e^{5x}$
- d. $e^{\frac{1}{2}x}$
- e. e^{ax}
- f. e^{-kx}
- g. $-e^{-\pi x}$
- h. $e^{\frac{1}{e}x}$

2. Find the derivative of:

- a. e^{2x-1}
- b. e^{1-x}
- c. $3e^{-3x+4}$
- d. $2e^{\frac{1}{2}x+4}$
- e. e^{px+q}
- f. $e^{2x} - e^{-3x}$
- g. $\frac{e^x - e^{-x}}{2}$
- h. $\frac{e^{ax}}{a} + \frac{e^{-bx}}{b}$

3. Write as a power of e and then differentiate:

- a. $(e^x)^2$
- b. $\frac{1}{e^{3x}}$
- c. $\sqrt{e^x}$
- d. $\frac{1}{(\sqrt{e^x})^3}$

4. Use the product and chain rules as appropriate to differentiate:

- a. e^{x^2}
- b. e^{1-x^2}
- c. e^{x^2+2x}
- d. e^{6+x-x^2}
- e. $\frac{1}{2}e^{2x^2-2x+1}$
- f. xe^x
- g. xe^{-x}
- h. $(x-1)e^x$
- i. $(x+1)e^{3x-4}$
- j. xe^{x^2+1}
- k. $(x^2+5x-5)e^x$
- l. $(x^2-x^3)e^{-x}$

5. a. Find the gradient of the curve $y = e^{5x}$ at the point $A(a, e^{5a})$, and show that the gradient is 5 times the height

b. Find the gradient of the curve $y = be^{-3x}$ at the point $A(a, be^{-3a})$, and show that the gradient is -3 times the height.

c. Find the gradient of the curve $y = be^{kx}$ at the point $A(a, be^{ka})$, and show that the gradient is k times the height.

6. Use the product, quotient, chain and log rules as appropriate to differentiate:

- a. $(x^2 - x)e^{2x-1}$
- b. $\ln(1 - e^x)$
- c. $\ln(e^x + x)$
- d. $e^x \ln x$
- e. $\ln(e^x + e^{-x})$
- f. $\frac{e^x}{x}$
- g. $\frac{x+1}{e^x}$
- h. $\frac{e^x+1}{e^x-1}$
- i. $\frac{e^x}{(x+1)^2}$
- j. $\frac{e^x - e^{-x}}{e^x + e^{-x}}$

7. Use the identity $a = e^{\ln a}$ to write each expression as a power of e . Thus differentiate it.

- a. 2^x
- b. 10^x
- c. π^x
- d. a^x
- e. 2^{3x-1}
- f. 5^{2-x}
- g. a^{x-2}
- h. a^{bx+c}
- i. $x2^x$
- j. 3^{x^3-3x}

8. Simplify then differentiate:

- a. $e^{\ln x}$
- b. $e^{-\ln x}$
- c. $\ln(e^{-x})$
- d. $\ln(e^{3x-7})$

9. a. Show that $Ae^0, Ae^{-1}, Ae^{-2}, \dots$ forms a GP and find its limiting sum.
 b. Show that for the AP $u_1 = a, u_2 = a + d, u_3 = a + 2d, \dots$. The sequence $e^{u_1}, e^{u_2}, e^{u_3}, \dots$ is a GP.
 c. Conversely, show that for the GP $u_1 = a, u_2 = ar, u_3 = ar^2, \dots$, the sequence $\ln u_1, \ln u_2, \ln u_3, \dots$ is an AP.
10. a. Show that both $y = e^{-x}$ and $y = xe^{-x}$ are solutions of $y'' + 2y' + y = 0$.
 b. Show that $y = e^{-3x} + x - 1$ is a solution of the differential equation $y'' + 2y' - 3y = 5 - 3x$.
 c. Find the value of λ that make $y = 5e^{\lambda x}$ is a solution of:
 i. $y'' + 3y' - 10y = 0$
 ii. $y'' + y' - y = 0$
11. a. Show that $y = Ae^{kx}$ is a solution of:
 i. $y' = ky$
 ii. $y'' - k^2y = 0$
 b. Show that $y = Ae^{kx} + C$ is a solution of $\frac{dy}{dx} = k(y - C)$
 c. Show that $y = (Ax + B)e^{3x}$ is a solution of $y'' - 6y' + 9y = 0$
12. Differentiate:
 a. $e^{\frac{1}{x}}$
 b. $e^{x+\ln x}$
 c. $3^{-\frac{1}{x}}$
13. a. Find the x -coordinates of the stationary points of $y = xe^{-x^2}$.
 b. Determine where the second derivative of $y = x^2e^x$ changes sign.
 c. Let $f(x) = xe^x$ and $g(x) = xe^{-x}$. Find $f'(x)g'(x)$.
14. We define two new functions $\cosh x = \frac{e^x + e^{-x}}{2}$ and $\sinh x = \frac{e^x - e^{-x}}{2}$.
 a. Show that each of these functions is the derivative of the other.
 b. Show that both functions are solutions of the differential equation $y'' - y = 0$.
 c. Show that $\cosh^2 x - \sinh^2 x = 1$.
15. Differentiate, either by expressing as a power of e , or by taking logs of both sides and using implicit differentiation:
 a. $y = x^x$
 b. $y = x^{2-x}$
 c. $y = x^{\ln x}$
 d. $y = x^{\frac{1}{\ln x}}$
16. a. Find the possible values of λ that make $y = e^{\lambda x}$ a solution of $ay'' + by' + cy = 0$
 b. When will there be no real solution for λ .

Answer

1. a. $2e^{2x}$; b. $-3e^{-3x}$; c. $-5e^{5x}$; d. $\frac{1}{2}e^{\frac{1}{2}x}$; e. ae^{ax} ; f. $-ke^{-kx}$; g. $\pi e^{-\pi x}$; h. $-\frac{1}{c}e^{-\frac{1}{c}x}$
2. a. $2e^{2x-1}$; b. $-e^{1-x}$; c. $-9e^{-3x+4}$; d. $e^{\frac{1}{2}x+4}$; e. pe^{px+q} ; f. $2e^{2x} + 3e^{-3x}$; g. $\frac{(e^x + e^{-x})}{2}$; h. $e^{ax} - e^{-bx}$
3. a. $2e^{2x}$; b. $-\frac{3}{e^{3x}}$; c. $\frac{1}{2}e^{\frac{x}{2}}$; d. $-\frac{3}{2}e^{-\frac{3x}{2}}$
4. a. $2xe^{x^2}$; b. $-2xe^{1-x^2}$; c. $2(x+1)e^{x^2+2x}$; d. $(1-2x)e^{6+x-x^2}$; e. $(3x-1)e^{3x^2-2x+1}$; f. $(x+1)e^x$; g. $(1-x)e^{-x}$; h. xe^x ; i. $(3x+4)e^{3x-4}$; j. $(2x^2+1)e^{x^2+1}$; k. $(x^2+7x)e^x$; l. $(x^3-4x^2+2x)e^{-x}$
6. a. $(2x^2-1)e^{2x-1}$; b. $-\frac{e^x}{1-e^x}$; c. $\frac{e^{x+1}}{e^x+x}$; d. $e^x(\ln x + \frac{1}{x})$; e. $\frac{e^x - e^{-x}}{e^x + e^{-x}}$; f. $\frac{(x-1)e^x}{x^2}$; g. $-xe^{-x}$; h. $-\frac{2e^x}{(e^x-1)^2}$; i. $\frac{(x-1)e^x}{(x+1)^3}$; j. $\frac{4}{(e^x + e^{-x})^2}$
7. a. $2^x \ln 2$; b. $10^x \ln 10$; c. $\pi^x \ln \pi$; d. $a^x \ln a$; e. $2^{3x-1} 3 \ln 2$; f. $-5^{2-x} \ln 5$; g. $a^{x-2} \ln a$; h. $a^{bx+c} b \ln a$; i. $(x \ln 2 + 1)2^x$; j. $(3x^2-3)3^{x^3-3x} \ln 3 = (x^2-1)3^{x^3-3x+1} \ln 3$
8. a. 1; b. $-\frac{1}{x^2}$; c. -1; d. 3;
9. a. $\frac{Ae}{e-1}$; b. The common ratio is e^d ;
10. c. i. -5 or 2, ii. $-\frac{1+\sqrt{5}}{2}$ or $-\frac{1-\sqrt{5}}{2}$
12. a. $-\frac{1}{x^2}e^{\frac{1}{x}}$; b. $(x+1)e^x$; c. $\frac{\ln 3}{x^2}3^{-\frac{1}{x}}$
13. a. $-\frac{1}{\sqrt{2}}$ or $\frac{1}{\sqrt{2}}$; b. $-2 - \sqrt{2}$ and $-2 + \sqrt{2}$; c. $1 - x^2$
15. a. $y = e^x \ln x$, so $y' = (\ln x + 1)x^x$; b. $y = e^{(2-x)\ln x}$, so $y' = (\frac{2}{x} - 1 - \ln x)x^{2-x}$; c. $y = e^{(\ln x)^2}$, so $y' = \frac{(2x \ln x \ln x)}{x}$; d. $y = e^{\ln x \times \frac{1}{\ln x}}$, so $y' = 0$
16. a. $\frac{-b + \sqrt{b^2 - 4ac}}{2a}$ or $\frac{-b - \sqrt{b^2 - 4ac}}{2a}$; b. when $b^2 - 4ac < 0$